Varshitha Dugyala

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Summary

Self-driven and detail-oriented undergraduate interested in using technology to solve real-world issues. Strong communication, teamwork, and time management skills with a proven track record of balancing academics and project work. Looking to apply a positive, curious, and proactive mindset to a dynamic team setting.

Education

Vasavi College of Engineering Hyderabad, India
Bachelor of Engineering - CSE (AIML); CGPA: 9.04 *2022 – Present*

Projects

Driver Drowsiness Detection System | *Python, OpenCV, TensorFlow, Keras* Sep 2024

- Developed a real-time driver drowsiness detection system using CNN-based eye state classification, yawn detection, and head pose estimation with alert triggers to prevent accidents.
- Implemented the Eye Aspect Ratio (EAR) algorithm to quantify eye openness, ensuring high sensitivity to micro-sleep events while filtering out normal blinking.
- Technologies: Python, OpenCV, Dlib, TensorFlow, Keras, NumPy, SciPy, Pygame.

Text to 3D Scene Generation | *Python, NLP* July 2025

- Created a system to generate 3D scenes from natural language using NLP and Transformer models by mapping text to scene parameters.
- Engineered a semantic mapping module that translates linguistic attributes into spatial coordinates, allowing for the automated rendering of complex scene compositions.
- Technologies: Python, NLP.

Online Quizzes and Courses Platform | *MERN Stack, Gemini API* Mar 2024

- Developed a full-stack platform with user login, demo videos, payment integration, and an AI-powered quiz generator for personalized assessments.
- Integrated dynamic prompt engineering to tailor Gemini's output to specific difficulty levels, ensuring generated content remains contextually relevant to the user's selected curriculum.
- Technologies: React, Node.js, MongoDB, CSS, Gemini API.

Technical Skills

Programming Languages: C, C++, Python, Java, JavaScript
Database Management: MySQL, MongoDB, Supabase, Firebase
Web Applications: HTML, CSS, React, Node JS
Frameworks: Git, Docker
Agentic & AI/ML Frameworks: LangGraph, LangChain, Pytorch, numpy, pandas
Software Engineering: SDLC Lifecycle, DBMS, OOPS, OS, Computer Networks

Achievements

NPTEL 'Joy Of Computing Using Python': Elite+Silver (Apr 2024)
NPTEL 'Programming in C++': Elite (July 2023)

Co-Curricular Activities

NSS: Worked in NSS and Organized several events in the college campus (2023-2024)
Toastmasters Club: Member of an Elite Communications Club in Vasavi (July 2023)